

DATA X LABS – INTERNSHIP (2026)

EXPLORATORY DATA ANALYSIS (EDA)

NETFLIX MOVIES AND TV SHOWS DATASET

DATA ANALYST INTERNSHIP – TASK 5

PROJECT REPORT

Submitted By:

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Tools Used:

- Python
- Pandas
- Matplotlib
- Seaborn
- Google Colab

Dataset Used:

Netflix Movies and TV Shows Dataset

Objective:

To perform Exploratory Data Analysis (EDA) on the Netflix dataset and identify meaningful trends, patterns, and insights using statistical analysis and visualizations.

Dataset Overview

The Netflix dataset contains information about movies and TV shows available on Netflix.

Dataset Information:

Total Records: 8,807

Total Columns: 12

Columns Included:

- Content Type
- Title
- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre
- Description

DATA PREPARATION

Before analysis, the dataset was cleaned and preprocessed. Cleaning Activities Performed

- Removed duplicate records
- Handled missing values
- Standardized column names
- Corrected data formats
- Verified data types
- Prepared dataset for analysis

```
[ ] df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column              Non-Null Count  Dtype
---  -
0   show_id              8807 non-null   object
1   type                 8807 non-null   object
2   title                8807 non-null   object
3   director             8807 non-null   object
4   cast                 8807 non-null   object
5   country              8807 non-null   object
6   date_added           8709 non-null   object
7   release_year         8807 non-null   int64
8   rating               8807 non-null   object
9   duration             8804 non-null   object
10  listed_in            8807 non-null   object
11  description           8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Fig 1: Dataset Structure Information

```
[ ] df.describe(include="all")

***
   show_id  type  title  director  cast  country  date_added  release_year  rating  duration  listed_in  description
count  8807  8807  8807  8807  8807  8807  8709  8807.000000  8807  8804  8807  8807
unique  8807  2  8807  4529  7693  749  1699  NaN  18  220  514  8775
top  s8807  Movie  Zubaan  Unkown  Unkown  United States  2020-01-01  NaN  TV-MA  1 Season  Dramas, International Movies  Paranormal activity at a lush, abandoned prope...
freq  1  6131  1  2634  825  2818  109  NaN  3207  1793  362  4
mean  NaN  NaN  NaN  NaN  NaN  NaN  NaN  2014.180198  NaN  NaN  NaN  NaN
std  NaN  NaN  NaN  NaN  NaN  NaN  NaN  8.819312  NaN  NaN  NaN  NaN
min  NaN  NaN  NaN  NaN  NaN  NaN  NaN  1925.000000  NaN  NaN  NaN  NaN
25%  NaN  NaN  NaN  NaN  NaN  NaN  NaN  2013.000000  NaN  NaN  NaN  NaN
50%  NaN  NaN  NaN  NaN  NaN  NaN  NaN  2017.000000  NaN  NaN  NaN  NaN
75%  NaN  NaN  NaN  NaN  NaN  NaN  NaN  2019.000000  NaN  NaN  NaN  NaN
max  NaN  NaN  NaN  NaN  NaN  NaN  NaN  2021.000000  NaN  NaN  NaN  NaN
```

Fig 2: Summary Statistics of the Dataset

```
MISSING VALUES

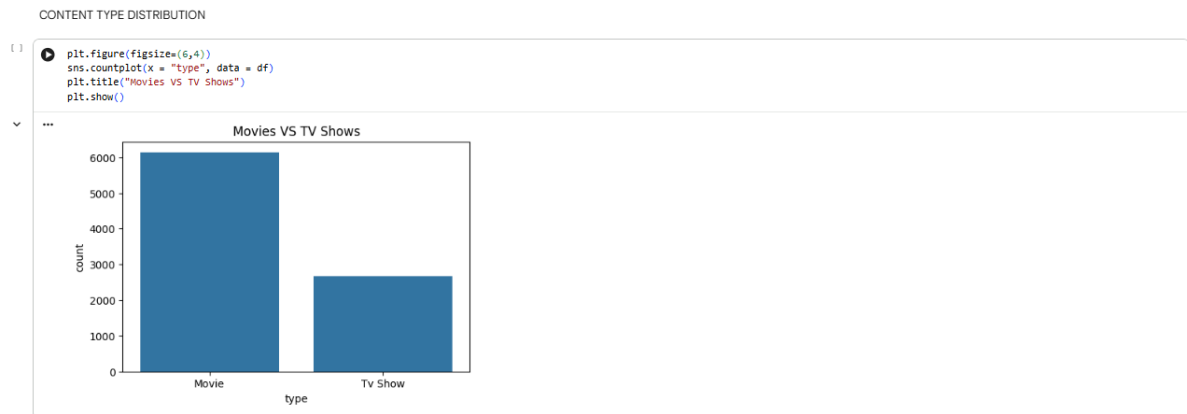
[ ] df.isnull().sum()

***
   show_id  0
   type    0
   title   0
   director 0
   cast     0
   country  0
   date_added  98
   release_year  0
   rating      0
   duration    3
   listed_in   0
   description  0
dtype: int64
```

Fig 3: Missing Values Analysis

Visualization 1:

MOVIES VS TV SHOWS

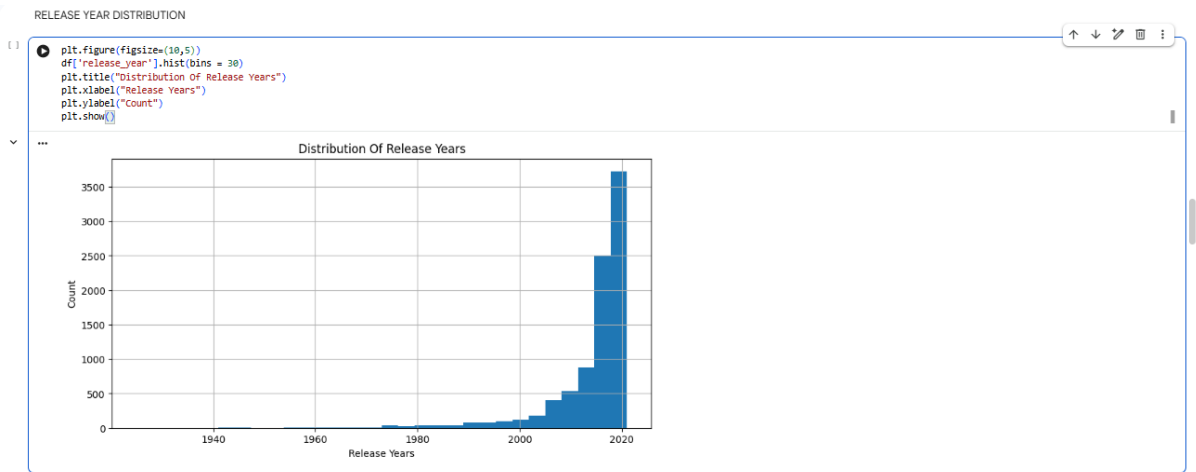


Observation:

- Movies represent a larger share of Netflix content compared to TV Shows.
- Netflix focuses heavily on movie content.

Visualization 2:

DISTRIBUTION OF RELEASE YEARS

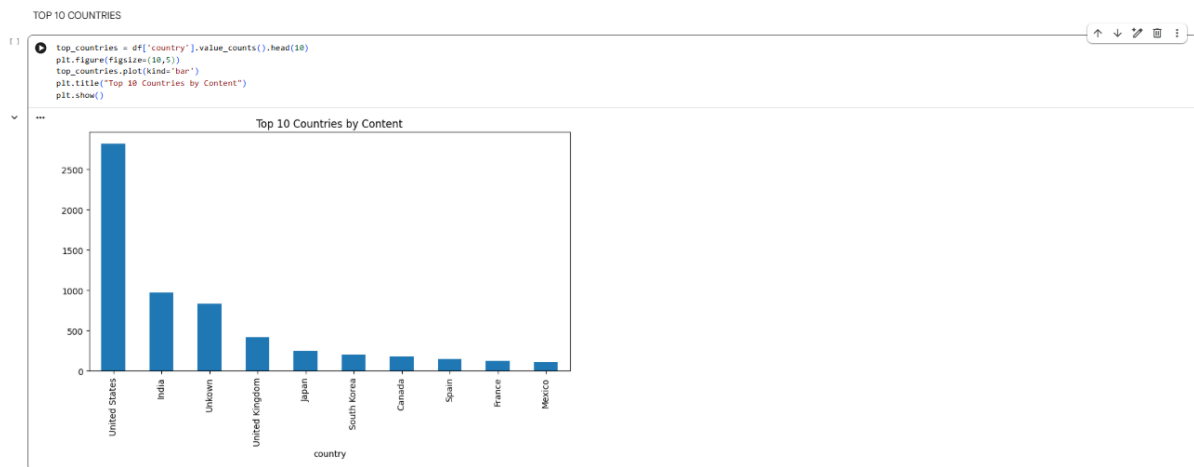


Observation:

- Most content was released in recent years.
- Content production increased significantly after 2015.

Visualization 3:

TOP 10 COUNTRIES BY CONTENT

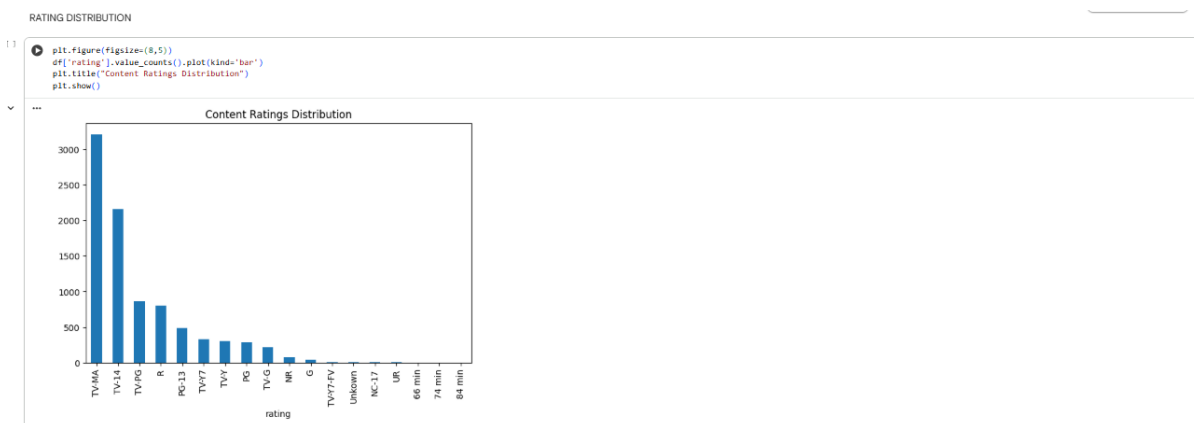


Observation:

- United States contributes the highest number of titles.
- Netflix contains content from many countries.

Visualization 4:

CONTENT RATINGS DISTRIBUTION

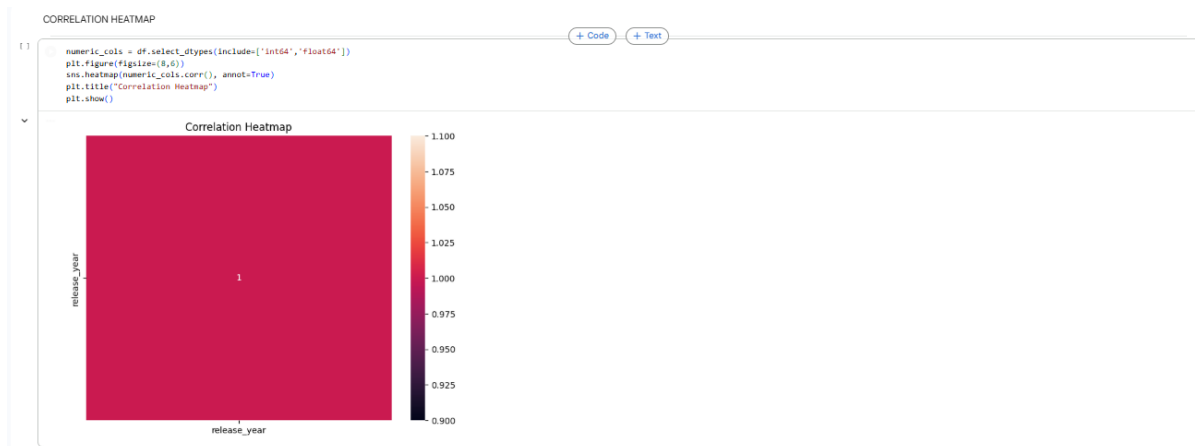


Observation:

- TV-MA and TV-14 are among the most common ratings.
- Netflix provides content for various audience groups.

Visualization 5:

CORRELATION HEATMAP

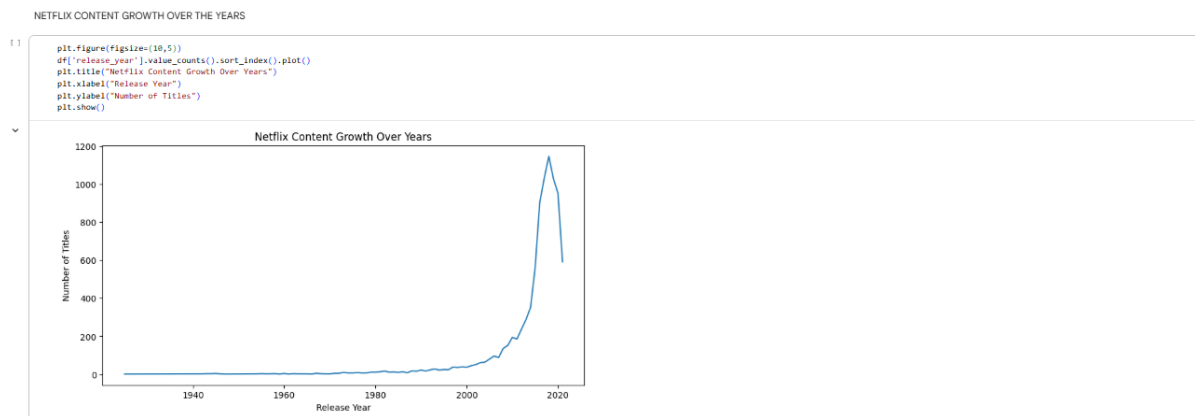


Observation:

- Limited numerical features are available in the dataset.
- No strong correlation analysis can be performed.

Visualization 6:

NETFLIX CONTENT GROWTH OVER THE YEARS



Observation:

- Netflix content has grown rapidly over the years.
- Significant expansion occurred after 2015.

Key Findings & Conclusion:

KEY FINDINGS:

1. Movies dominate the Netflix content library.
2. Content production increased rapidly after 2015.
3. The United States contributes the highest amount of content.
4. TV-MA and TV-14 are the most common ratings.
5. Netflix has expanded globally with content from multiple countries.
6. Recent years show significant growth in content availability.

CONCLUSION:

Exploratory Data Analysis helped identify important trends and patterns within the Netflix dataset. The analysis revealed content distribution, growth trends, country contributions, and audience-targeting patterns. The project demonstrates how EDA can be used to transform raw data into meaningful business insights.

SKILLS LEARNED:

- Exploratory Data Analysis
- Data Visualization
- Statistical Analysis
- Data Interpretation
- Trend Analysis
- Python
- Pandas
- Matplotlib
- Seaborn